Case Study 16: Multi-Day Water Tank Simulation

1. Problem Statement

Process dynamic daily inflow and outflow volumes, maintain current water storage levels, detect overflow and shortage conditions, and determine the exact day when the tank level first reaches or drops below a critical threshold.

2. Objectives

- Simulate multi-day water inventory using cumulative storage tracking.
- Enforce physical tank boundaries (capacity limit and non-negative minimum).
- Identify critical alert conditions including overflow wastage and shortage events.

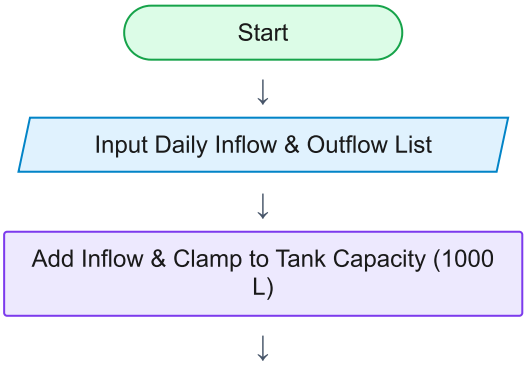
3. Assumptions

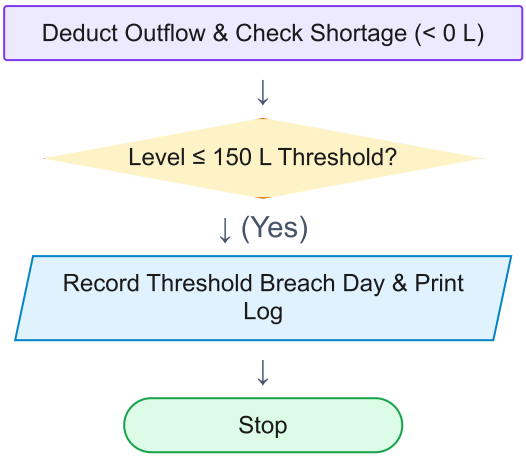
- Maximum Tank Capacity = **1,000 Liters**.
- Initial Storage Level = **500 Liters**.
- Critical Low Threshold = **150 Liters**.
- Inflow occurs before daily consumption (outflow).

4. Algorithm

1. Start.
2. Initialize `current_level = 500`, `capacity = 1000`, and `threshold = 150`.
3. Iterate through daily `(inflow, outflow)` records for `day = 1, 2, ... N`.
4. Calculate intermediate level after inflow: `level = level + inflow`.
5. Check overflow: If `level > capacity`, record overflow volume and set `level = capacity`.
6. Subtract outflow: `level = level - outflow`.
7. Check shortage: If `level < 0`, record shortage volume and reset `level = 0`.
8. Check critical threshold: Flag the first day `level <= threshold`.
9. Display daily log and multi-day status summary.
10. Stop.

5. Flowchart





6. Pseudocode

```
      START capacity = 1000, current = 500, threshold = 150 first_threshold_day =
-1 FOR day FROM 1 TO length(daily_data) DO inflow = daily_data[day].inflow outflow
= daily_data[day].outflow current = current + inflow IF current > capacity THEN
overflow = current - capacity current = capacity ELSE overflow = 0 ENDIF current =
current - outflow IF current < 0 THEN shortage = ABS(current) current = 0 ELSE
shortage = 0 ENDIF IF current <= threshold AND first_threshold_day == -1 THEN
first_threshold_day = day ENDIF PRINT day, inflow, outflow, current, overflow,
shortage ENDFOR STOP
```

7. Complete Python Source Code

```
      # Multi-Day Water Tank Simulation def run_water_tank_simulation(daily_rates):
capacity = 1000.0 current_level = 500.0 critical_threshold = 150.0
first_critical_day = None total_overflow = 0.0 total_shortage = 0.0 print("=" * 70)
print(" MULTI-DAY WATER TANK SIMULATION LOG") print("=" * 70) print(f"{'Day':<6}
{'In (L)':<10}{'Out (L)':<10}{'Level (L)':<12}{'Overflow':<12}{'Shortage':<10}")
print("-" * 70) for day, (inflow, outflow) in enumerate(daily_rates, 1): overflow =
0.0 shortage = 0.0 # Step 1: Add Inflow current_level += inflow if current_level >
capacity: overflow = current_level - capacity total_overflow += overflow
current_level = capacity # Step 2: Deduct Outflow current_level -= outflow if
current_level < 0: shortage = abs(current_level) total_shortage += shortage
current_level = 0.0 # Step 3: Check critical threshold if current_level <=
critical_threshold and first_critical_day is None: first_critical_day = day
print(f"{day:<6}{inflow:<10.1f}{outflow:<10.1f}{current_level:<12.1f}{overflow:
<12.1f}{shortage:<10.1f}") print("-" * 70) print(f"Total Volume Overflowed :
{total_overflow:.1f} Liters") print(f"Total Shortage Volume : {total_shortage:.1f}
Liters") if first_critical_day: print(f"Critical Threshold Breach: Day
{first_critical_day} (Level ≤ {critical_threshold} L)") else: print(f"Critical
Threshold Breach: None (Level maintained > {critical_threshold} L)") print("=" *
70) # Multi-Day Test Data: (Inflow, Outflow) test_schedule = [ (200, 300), # Day 1:
Level 400 (500, 200), # Day 2: Level 700 (450, 100), # Day 3: Level 1000 (50L
overflow) (100, 600), # Day 4: Level 500 (50, 400), # Day 5: Level 150 (Breaches
threshold) (50, 300) # Day 6: Level 0 (100L shortage) ]
run_water_tank_simulation(test_schedule)
```

8. Test Case and Expected Output

Simulation Parameters: Initial Level = 500 L, Capacity = 1000 L, Critical Threshold = 150 L

```
      ====================================================================== MULTI-
DAY WATER TANK SIMULATION LOG
====================================================================== Day In (L)
Out (L) Level (L) Overflow Shortage ------------------------------------------------
---------------------- 1 200.0 300.0 400.0 0.0 0.0 2 500.0 200.0 700.0 0.0 0.0 3
450.0 100.0 1000.0 50.0 0.0 4 100.0 600.0 500.0 0.0 0.0 5 50.0 400.0 150.0 0.0 0.0
6 50.0 300.0 0.0 0.0 100.0 ------------------------------------------------------------
-------------- Total Volume Overflowed : 50.0 Liters Total Shortage Volume : 100.0
Liters Critical Threshold Breach: Day 5 (Level ≤ 150.0 L)
======================================================================
```

9. Conclusion

The Multi-Day Water Tank Simulation model accurately tracks dynamic fluid levels, detects capacity overflow and deficit shortage events, and identifies threshold breaches to facilitate proactive water management.